IT Career Course Directory — Certificate-First Edition

Every course below leads to SOME certificate/badge (free preferred). Tools/accounts (GitHub, labs, VMs) are listed too since they're needed, but are not courses themselves. Italicized notes flag what changed and why.

SETUP (tools/accounts — not courses)

Program / Course	Where to Get It
GitHub account + portfolio repo	github.com
VirtualBox (free VM software)	virtualbox.org
VMware Workstation Player (free VM software)	vmware.com/products/workstation-player.html

CORE FOUNDATIONS — MONTHS 1-3

Program / Course	Where to Get It
CompTIA A+ Core 1 (220-1201) — free study	professormesser.com
CompTIA A+ Core 2 (220-1202) — free study	professormesser.com
A+ official exam (paid, ~$253-274 each) — issues the actual certification	comptia.org
Google IT Support Professional Certificate — issues a real Professional Certificate	coursera.org/professional-certificates/google-it-support
Cisco NetAcad — "Networking Basics" (22 h)	netacad.com
Cisco NetAcad — "Getting Started with Cisco Packet Tracer" (2 h) — DONE	netacad.com
Cisco NetAcad — "Networking Devices and Initial Configuration" (22 h)	netacad.com
Cisco NetAcad — "Network Addressing and Basic Troubleshooting" (14 h)	netacad.com
Cisco NetAcad — "Introduction to Cybersecurity" (6 h)	netacad.com
Cisco NetAcad — "Operating Systems Basics" (12 h)	netacad.com
Cisco NetAcad — "Operating Systems Support" (10 h)	netacad.com
Microsoft Learn — "MD-100: Install the Windows client" (free study)	learn.microsoft.com/training
Microsoft Learn — "MD-100: Support the Windows client environment" (free study)	learn.microsoft.com/training
Microsoft Learn — "Secure access to resources by using Microsoft Entra" (free study)	learn.microsoft.com/training
Free Azure account (auto-creates an Entra ID tenant)	azure.microsoft.com/free
Microsoft Applied Skills — "Get started with identities and access using Microsoft Entra"	learn.microsoft.com/credentials/applied-skills
Oracle MyLearn — "Become an OCI Foundations Associate 2026"	mylearn.oracle.com
freeCodeCamp — "Relational Database Certification" (100% free, verified certificate)	freecodecamp.org
GitHub Skills — "Introduction to GitHub"	skills.github.com

- *Completed Oct 4, 2026; certificate issued.*
- *Free, self-paced. Check the course page for the badge it issues.*
- *Free, self-paced. Check the course page for the badge it issues.*
- *Replaces a prior Juniper slot that had no standalone free certificate.*
- *Free, self-paced.*
- *Free, self-paced.*
- *Free training only — no certificate on its own. Feeds into the Applied Skills credential below.*

■ *Free training only. For an actual certification, target MD-102 instead — confirmed $165 USD.*
■ *Free training only. Feeds into the Applied Skills credential below.*
■ *Tool/account, not a course — required to complete the Applied Skills credential below.*
■ *THE free credential that the three Microsoft Learn items above build toward.*
■ *NEW — replaces PlanetScale's "MySQL for Developers," which teaches similar material but issues no certificate.*
■ *Hands-on practice tool — issues profile achievement badges, not a formal certificate. Optional paid GitHub Foundations certification (~$99) exists if you want a formal credential.*

SECURITY, IDENTITY & CLOUD — MONTHS 4-6

Program / Course	Where to Get It
Fortinet — "Fortinet Certified Fundamentals (FCF)"	training.fortinet.com
Palo Alto Networks Beacon — free "Cybersecurity Foundation" course	paloaltonetworks.com/academy
ISC2 — "Certified in Cybersecurity (CC)"	isc2.org/Certifications/CC
Microsoft — "AZ-900: Azure Fundamentals"	learn.microsoft.com/credentials/certifications/azure-fundamentals
Microsoft Applied Skills — Azure task-based credentials	learn.microsoft.com/credentials/applied-skills
AWS Skill Builder — "Cloud Practitioner Essentials"	explore.skillbuilder.aws
Google Cloud Skills Boost — "Cloud Digital Leader"	cloudskillsboost.google
MongoDB University — "Introduction to MongoDB"	learn.mongodb.com
Redis University — "Get Started with Redis"	university.redis.io

■ *NEW — replaces Cyberpedia, which is reference content with no certificate.*
■ *Issues a free digital course-completion certificate, separate from the paid AWS Certification exam.*
■ *Issues a free Skill Badge on completion.*

TRACK A: NETWORK / SYSADMIN — MONTHS 7-9

Program / Course	Where to Get It
Cisco NetAcad — "Network Support and Security" (12 h)	netacad.com
Juniper Open Learning — "JNCIA-Junos" (elective)	juniper.net/us/en/training.html
VMware Hands-on Labs — ESXi & vSphere (hands-on tool)	vmware.com/try-vmware/hands-on-labs
Cisco CCST / CCNA exam (paid) — issues real certification	cisco.com/certifications
CompTIA Network+ exam (paid, ~$399) — issues real certification	comptia.org

■ *CORRECTION: Cisco's CCNA courses are instructor-led only (class code needed), so they are not free self-paced. This free course replaces CCNA: Introduction to Networks.*
■ *Training is free but issues no standalone certificate of its own — only the paid JNCIA exam does.*
■ *No certificate offered — kept purely for hands-on virtualization practice.*

TRACK B: CLOUD / DEVOPS — MONTHS 7-9

Program / Course	Where to Get It
IBM SkillsBuild — "Containers & Kubernetes" fundamentals (free badge)	skillsbuild.org
Linux Foundation — "Introduction to Kubernetes (LFS158)" via edX	training.linuxfoundation.org
HashiCorp Developer — Terraform "Get Started" tutorials (free study)	developer.hashicorp.com/terraform/tutorials
Linux Foundation — "Introduction to Linux (LFS101)"	training.linuxfoundation.org/training/introduction-to-linux
GitHub Skills — "GitHub Actions: Hello World" (hands-on practice tool)	skills.github.com

■ *NEW — replaces Docker's own "Get Started" guide, which has no certificate mechanism.*
■ *NEW — replaces kubernetes.io's own tutorials, which issue no certificate. Free honor-code certificate available on edX.*

- *Real credential is HashiCorp Certified: Terraform Associate — a relatively low-cost paid exam (~$70.50).*
- *Free honor-code certificate available.*
- *No formal certificate — practice only.*

TRACK C: DATABASE — MONTHS 7-9

Program / Course	Where to Get It
freeCodeCamp — "Relational Database Certification" (deeper practice)	freecodecamp.org
MongoDB University — "Aggregation Framework"	learn.mongodb.com
Couchbase Academy — free NoSQL courses	couchbase.com/academy
DataStax Academy — Cassandra Fundamentals	datastax.com

- *Likely issues a completion badge — exact current status not independently reverified this round.*

TRACK D: SECURITY & SOC — MONTHS 7-9

Program / Course	Where to Get It
Cisco NetAcad — "Network Defense" (27 h)	netacad.com
Cisco NetAcad — "Cyber Threat Management" (16 h)	netacad.com
Cisco NetAcad — "Data and Tools for Defense Analysts" (6 h)	netacad.com
Cisco NetAcad — "Introduction to Splunk" (1 h)	netacad.com
Cisco NetAcad — "SOC Essentials: Investigating with Splunk" (3 h)	netacad.com
Cisco NetAcad — "Ethical Hacker" (70 h) — optional	netacad.com
Wazuh (self-hosted SOC lab tool)	wazuh.com
Security Onion (self-hosted SOC lab tool)	securityonionsolutions.com
CompTIA Security+ exam (paid, ~$439) — issues real certification	comptia.org

- *Free, self-paced.*
- *Free, self-paced.*
- *Free, self-paced.*
- *Replaces the earlier vague Splunk Education entry. Free, self-paced.*
- *Free, self-paced.*
- *Free, self-paced. Long; only for offensive-security depth.*
- *Not a course — software for your own capstone lab.*
- *Not a course — software for your own capstone lab.*

TRACK E: AI / ML — MONTHS 7-9

Program / Course	Where to Get It
Google Cloud — "Generative AI Fundamentals" badge	cloudskillsboost.google
IBM SkillsBuild — "AI Fundamentals: Foundations for Understanding AI" (9 h)	skillsbuild.org
IBM SkillsBuild — "AI Fundamentals: Language and Vision in AI" (10 h)	skillsbuild.org
Cisco NetAcad — "Core Generative AI Skills" (7 h)	netacad.com
Cisco NetAcad — "Prompt Like an Engineer" (9 h)	netacad.com
Hugging Face — "Agents Course" (free certificate)	huggingface.co/agents-course
NVIDIA Deep Learning Institute — free self-paced courses	nvidia.com/en-us/training
AWS — AI Practitioner learning path	aws.amazon.com/certification/certified-ai-practitioner
Microsoft — "AI-901: Azure AI Fundamentals"	learn.microsoft.com/credentials/certifications/exams/ai-901

- *Free, self-paced.*
- *Free, self-paced.*
- *Free, self-paced.*
- *Free, self-paced.*
- *Several issue a certificate of competency on passing the in-course assessment — confirm per course.*
- *Issues a free digital course-completion certificate, separate from the paid AWS exam.*

ELECTIVE: ENDPOINT & MOBILE DEVICE MANAGEMENT — MONTHS 10-12

Program / Course	Where to Get It
Microsoft Applied Skills — "Defend against cyberthreats with Microsoft Defender XDR"	learn.microsoft.com/credentials/applied-skills/defend-against-cyberthreats-with-microsoft-defender-xdr
Microsoft Learn — "App management using Microsoft Endpoint Manager" (MAM, BYOD, app protection)	learn.microsoft.com/training/modules/app-management-using-microsoft-endpoint-manager
Jamf 100 Course — Apple device management (MDM) fundamentals	jamf.com/training/100
Android Enterprise Academy — Android device fleet management	androidenterprise.training
Cisco NetAcad — "Endpoint Security" (27 h)	netacad.com

- *Free hands-on credential.*
- *Free training — no standalone certificate; supports the Intune credentials above.*
- *Training is free. The optional Jamf Certified Associate exam is a separate ~$100 paid credential.*
- *Free training AND free Android Enterprise Associate certification.*
- *Free, self-paced.*

ELECTIVE: VIRTUALIZATION, STORAGE & BACKUP — MONTHS 10-12

Program / Course	Where to Get It
NDG "Linux Unhatched" (8 h), then "Linux Essentials" (70 h) — on Cisco NetAcad	netacad.com
Red Hat — "OpenShift Virtualization Technical Overview (DO016)" (free on-demand video course)	redhat.com/en/services/training
Red Hat — "Containers, Kubernetes and Red Hat OpenShift Technical Overview (DO080)" (free course)	redhat.com/en/services/training
Veeam University Free — Veeam Backup & Replication / Veeam Data Platform training	veeam.com/university.html
Veeam Community Edition (free license, home-lab practice tool)	veeam.com
Dell Technologies — "Information Storage and Management Foundations" (SAN/storage credential)	dell.com

- *Both free, self-paced. Linux Essentials prepares for the LPI Linux Essentials exam, a separate paid credential.*
- *Find course code DO016 in the catalog. Free, no prerequisites. Certificate status not confirmed; the formal EX316 credential is paid (price not confirmed).*
- *Find course code DO080 in the catalog. Certificate status not confirmed.*
- *Issues a FREE certificate on course completion. Formal VMCE/VMCA exams need a paid instructor-led class.*
- *Lab tool, not a course. Search 'Veeam Backup & Replication Community Edition' on the site.*
- *No free SAN/storage certificate was found. This is the credible paid option; exam fee not confirmed — search the title on Dell's certification pages.*

ENTERPRISE TOOLS & JOB SEARCH — MONTHS 10-12

Program / Course	Where to Get It
Microsoft Learn — Intune device-management paths (free study)	learn.microsoft.com/training
Simplilearn SkillUp — "Introduction to ITIL Foundation 4"	simplilearn.com/free-itil-foundation-course
Atlassian University — Jira Service Management fundamentals	university.atlassian.com
Microsoft — "SC-300: Identity and Access Administrator" (optional, $165)	learn.microsoft.com/credentials/certifications/identity-and-access-administrator

- *Feeds toward Intune-related Applied Skills credentials and/or MD-102.*
- *Several free Atlassian University courses issue a completion certificate — confirm the specific course you take.*
- *Strong credential if targeting Entra ID/identity-focused roles specifically.*